# Confidence-Weighted Dual-Teacher Networks With Biased Contrastive Learning for Semi-Supervised Semantic Segmentation in Remote Sensing Images

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***Abstract*— Semantic segmentation of remote sensing images is vital in remote sensing technology. High-quality models for this task require a vast amount of images, and manual annotation is a process that is time-consuming and labor-intensive. Consequently, this has catalyzed the emergence of semi-supervised semantic segmentation methods. However, the complexity of foreground categories in remote sensing images poses a challenge to maintaining prediction consistency. Moreover, inherent characteristics such as intraclass variations and interclass similarities result in a certain degree of confusion among features of different classes in the feature space. This impacts the final classification results. To improve the model's consistency and optimize the classification of categories based on features, this article proposes a new semi-supervised semantic segmentation framework that combines consistency regularization and contrastive learning (CL). In terms of consistency regularization, the proposed method incorporates dual-teacher networks, introduces ClassMix for image augmentation, and uses confidence levels to integrate the predictions from these networks. By introducing perturbations at both the network and image levels, while simultaneously maintaining consistency, the predictive prowess and generalization ability of the model are enhanced. For CL, positive-unlabeled learning (PU-Learning) is used to improve the problem of mis-sampling when selecting features. At the same time, higher biased weights are allocated to more challenging negative samples, thereby elevating the complexity of feature learning and enhancing the discriminative capability of the final feature representation space. Our extensive experiments on the ISPRS Vaihingen dataset and the challenging iSAID dataset have served to underscore the superior performance of our approach.**

***Index Terms*— Consistency regularization, contrastive learning (CL), remote sensing, semantic segmentation, semi-supervised learning.**

## I. Introduction

SEMANTIC segmentation is a pivotal task in remote sensing image processing [1], [2], [3], [4], which entails the classification of individual pixels in an image into distinct categories. This technique has a wide range of applications, such as road extraction, urban planning, and land cover classification. However, successful execution of this task necessitates a significant number of images for model training and optimization, and the manual annotation of pixels can be a time-consuming and labor-intensive process. In recent years, semi-supervised semantic segmentation has emerged as a crucial research direction for reducing the cost of manual annotation. Semi-supervised learning model uses only a small fraction of the labeled dataset during the training process.

The state-of-the-art (SOTA) semi-supervised semantic segmentation methods can be categorized into these approaches: consistency regularization [5], contrastive learning (CL) [6], and a hybrid method that combines elements of the previous two [6], [7]. The consistency regularization-based method introduces perturbations at various levels during the training process of the same unlabeled image while maintaining output consistency, which involves ensuring the output remains stable despite changes induced by the perturbations. These perturbations can be categorized into multiple levels: image-level perturbations involve diverse image enhancements [8], [9], feature-level perturbations introduce noise [10], [11], and network-level perturbations add extra networks [12], [13]. Unlike the consistency regularization approach, CL follows a principle of emphasizing differences in the feature space among different categories while reducing similarities within the same category [14]. It can be observed that methods based on consistency regularization and CL operate at different levels. Therefore, some studies have combined these two approaches [6], [15] and achieved promising results.

However, compared with visual scenes, the foreground content of remote sensing images is more complex and diverse

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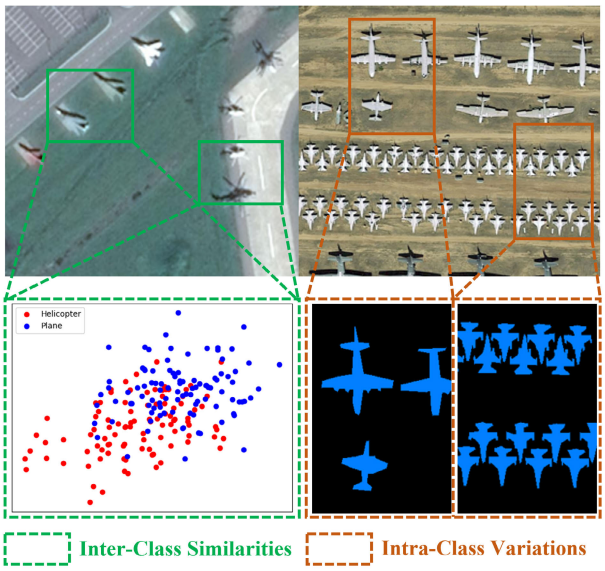



Fig. 1. Illustration of the potential challenges in remote sensing aerial imagery. 1) Interclass similarities. As shown in the green dashed box, the distribution of planes and helicopters in feature space is chaotic. 2) Intraclass variations. As shown in the brown dashed box, for the same category, such as plane, the differences are large.

than that of visual scenes. Applying methods designed for visual scenes can easily lead to poor consistency and misclassification of categories. Therefore, semi-supervised semantic segmentation in the remote sensing domain poses a challenging research task. As shown in Fig. 1, remote sensing images possess the following characteristics:

1) *Interclass Similarities:* Objects of different categories, such as airplanes and helicopters, can share substantial similarities.
2) *Intraclass Variations:* Objects of the same category often exhibit significant differences in features and scales.

In the field of remote sensing, the ICNet [16] improves the structure of CPS by incorporating a teacher network with better prediction performance, introducing perturbations at the network level, and using a moving average for parameter updates to improve the parameter update strategy of CPS and effectively prevent overfitting. However, the selection of the teacher network in ICNet is somewhat random and lacks a proper evaluation of the network, resulting in unreasonable perturbations. Compared with visual scene images, remote sensing images feature more complex foreground categories, making it more challenging to maintain consistent prediction results. Therefore, the existing consistency regularization models do not adapt well to remote sensing scenarios. In light of these challenges, this article aims to propose a new semi-supervised semantic segmentation framework that introduces more reasonable and powerful data and network perturbations to increase training difficulty, thereby achieving better training results and better adaptation to remote sensing scenarios.

Consistency regularization has improved model performance and robustness by introducing perturbations at the network and data levels. However, to construct a comprehensively excellent model, we need to optimize in three dimensions: network structure, feature representation, and image data. Considering the characteristics of remote sensing image features, optimizing at the feature level can better improve the performance of the model. Therefore, we introduce a CL strategy, which optimizes at the feature representation level, emphasizing the proximity of similar samples and the dissimilarity of dissimilar samples. This complements the optimization goal of consistency regularization, together constructing a powerful model framework optimized in all important dimensions.

As for CL, the current approach used in semi-supervised semantic segmentation models in the remote sensing domain [17] is based on Reco [6], which performs pixel-level CL. Reco selects query, positive key, and negative key from the generated pseudolabels. While it is simple to select query and positive key, the method of selecting negative key in these articles involves evenly sampling the remaining features. As query and positive key cannot be fully sampled, the positive key must exist in the remaining features. A simple uniform sampling of the remaining features is likely to include unsampled positive keys as negative keys. This will lead to contradictory CL strategies and have a significant impact on the final results. For remote sensing images, the significant intraclass variations cause the same class to be scattered in the feature space, which makes the correct sampling of positive and negative samples more difficult compared with visual scenes. Therefore, simple sampling criteria cannot meet the needs of remote sensing segmentation. In addition, the interclass similarities of remote sensing images make it difficult to accurately classify some samples. The existing methods based on CL loss use the InfoNCE [18] loss function to assign the same weight to the learning part of negative samples. For categories that are difficult to distinguish, this weighting method may lead to unsatisfactory learning results. Compared with visual scene images, there are more classes that are difficult to classify in remote sensing images due to the similarity between classes. Therefore, CL methods designed for visual scenes do not perform well when transferred to remote sensing images. Given the above, the two characteristics of remote sensing images result in poor performance when CL methods from natural scenes are transferred to remote sensing scenarios. For the sampling errors caused by the intraclass variations in remote sensing images, a more reasonable sampling method needs to be introduced to correctly divide positive and negative keys. As for the difficulty in classification caused by the interclass similarities of remote sensing images, simply using the standard InfoNCE loss does not effectively address this issue. Therefore, it is necessary to give higher attention to these hard-to-classify sample pairs during the learning process to better categorize them.

In this article, taking into account the issues present in current methods, we propose a novel semi-supervised semantic segmentation framework that combines consistency regularization and CL. To address the complexity of the foreground in remote sensing scenarios, it is necessary to reasonably increase the intensity of perturbation in consistency regularization. An additional teacher network based on mean teacher (MT) [5] is introduced to add perturbations at the network level, while also performing data augmentation at

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the image level. To make the perturbations more reasonable, labeled images are used to evaluate the teacher network, and higher confidence weightings are assigned to poorly performing images. These confidence weights are then used when merging the predicted maps of unlabeled images, appropriately increasing the prediction discrepancy between the student and teacher networks on complex categories during training, thus enhancing the model's generalization performance on difficult samples. When updating parameters, the parameters of the high-confidence teacher are adjusted while keeping the parameters of the low-confidence teacher unchanged, ensuring both diversity of parameters and smoothness of updates. This strategy appropriately enhances perturbations at both the image and network levels, thereby improving the model's predictive performance. In the CL module, to address the issue of feature mis-sampling caused by the dispersal of same-class features in feature space due to intraclass variations, PU learning is introduced to filter out truly difficult negative samples. Due to the interclass similarities of remote sensing images, classification problems with challenging category pairs are more common. Therefore, biased CL is used to increase the learning weight of these difficult negative samples in the image pixel space. This amplifies the feature learning intensity of hard-to-classify samples, ensuring higher classification accuracy for these samples.

The key contributions of this article can be summarized as follows.

1) We introduce a novel semi-supervised segmentation framework for remote sensing images. This framework integrates consistency regularization and CL, enhancing perturbations at both data and network levels, while also improving feature classification performance through CL. Moreover, this approach achieves SOTA performance on popular segmentation benchmarks.
2) A novel consistency regularization method is proposed based on MT [5]. Enhancing the disturbance at both the data and network levels makes it more difficult for the network to maintain consistency, increases the difficulty in training, and improves the generalization ability of complex images. In addition, using the confidence level to integrate predicted results from the teacher networks, the method can help the network learn more effectively from the unlabeled data, which can lead to better performance with limited labeled data.
3) Biased CL is used at the feature level to enhance the confidence of pseudolabels. PU-Learning is used for sample selection in CL, screening out true negative samples. During the process of CL, higher weights are assigned to more difficult negative samples to increase the difficulty in learning and improve classification performance in the final feature space.

The remainder of this article is organized as follows. Section II presents the primary related work, while Section III outlines the methodological specifics of this article. In Section IV, we introduce experimental details. In Section V, we analyze the effectiveness of our approach in conjunction with the experimental results. Finally, we conclude in Section VI.

## II. Related Works

### A. Supervised Semantic Segmentation

Semantic segmentation is a fundamental task in computer vision. Methods based on fully convolutional networks (FCNs) [19] can adapt to inputs of varying sizes while combining deconvolution and skip connections from different depth layers to improve the precision of the output results, laying the foundation for the application of deep learning in semantic segmentation.

Following the foundation laid by FCNs, UNet [20] introduces skip connections that link features from the encoder to the corresponding scale features of the decoder. This strategy provides a significant enhancement by preserving more detailed information during the segmentation process. Separately, SegNet [21] adopts a different approach, using pooling indices from the max-pooling layer for nonlinear upsampling. While SegNet also uses a skip connection concept, akin to UNet, its introduction of this new upsampling technique signifies an expansion to the existing practices in semantic segmentation, thereby enhancing its performance. PSPNet [22] parallels the work of SegNet by proposing a pyramid pooling module that performs pooling on specific processes at different dimensions. Complementing the approach of PSPNet, RefineNet [23] emphasizes on performing feature fusion using multidimensional, multiscale feature maps. It enhances the ability to capture object edges and details through a cascading approach. DeepLabV3+ [24] integrates the advancements from PSPNet and RefineNet using a spatial pyramid module and encoder–decoder structure for semantic segmentation, achieving excellent results and promoting the application of [25] and [26]. Based on FCN and combining the advantages of previous methods, FastFCN [27] proposes a joint pyramid upsampling (JPU) model, which achieves rapid upsampling of high-resolution feature maps.

### B. Semi-Supervised Learning

Semi-supervised learning uses only a small fraction of the labeled dataset during the training process. There are currently general learning strategies for semi-supervised learning: consistency-regularization-based semi-supervised learning, CL-based semi-supervised learning, and a hybrid method that combines elements of the previous two. Consistency-regularization-based semi-supervised learning emphasizes that the current model has consistency and stability in predicting the same data under various perturbations. Perturbations can be multifaceted, including at the data, feature, and network levels. The mainstream framework for this method has evolved from the most basic $\pi$-model [28] and temporal model [29] to MT. Most of the current method variations are based on MT, which optimizes a student network with the help of a teacher network, where the teacher network's parameters are the moving average of the student network's parameters. Based on this strategy, many semi-supervised methods for semantic segmentation have been developed. PS-MT [30]

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introduces T-VAT [10] at the feature level of the image, enhancing feature-level perturbations; CCT [31] performs data augmentation at the feature level, strengthening feature-level perturbations. CPS [13] uses networks with the same structure but different parameters for cross-constraints, thus achieving network perturbation; GCT [32] uses a collaboration strategy between teacher and student networks. ClassMix [8] and CutMix [9] perform data augmentation, outputting different views of the same image, and create data-level perturbations.

CL has excellent feature learning capabilities, learning the similarity of features within the same category and the dissimilarity of features between different categories for classification. CL is currently applied to self-supervised learning by many successful works [33], [34], [35]. There have been some works on semantic segmentation using CL [6], [36], [37], [38], [39]. Alonso et al. [36] build a memory bank with different category features to better perform feature learning. Lai et al. [40] adopt a self-supervised learning paradigm, selecting reliable pixel points from differently augmented images as positive and negative samples for CL.

The current research trend is to combine consistency-based and CL-based semi-supervised learning. Reco [6] first attempted by introducing a pixel-level CL task, while combining the MT framework to achieve better segmentation results. On the basis of reco, U2PL [15] combines memory bank to improve the long tail effect of data and enhance the disturbance at the data level. In addition, there are other types of methods as well. Huang et al. [41] use a self-training approach, measuring the confidence of pseudolabels during training to filter out more reliable pseudolabels for subsequent training. Vashishth et al. [42], building on GCNs, use the distance between two nodes as a measure of confidence, which reduces the influence of nodes with higher uncertainty in the process of information propagation.

### C. Positive-Unlabeled Learning

Positive-unlabeled learning (PU-Learning) is a machine learning methodology that can better filter positive and negative samples from the unlabeled sample space. This approach is particularly useful in tasks such as retrieval and outlier detection, where labeled negative samples might be scarce or not available at all [43], [44], [45]. Numerous research efforts have concentrated on diversifying loss functions to boost the efficacy of PU-Learning. Notably, the emphasis on zero-one losses is notably highlighted in the work of [44], [45], [46]. These studies argue that minimizing the zero-one loss is crucial in PU-Learning as it directly corresponds to the error rate, the most natural performance measure in classification tasks. They proposed various strategies to approximate the zero-one loss, which led to significant improvements in the accuracy of the resulting models. On the other hand, Chuang et al. [47] investigated the application of contrastive losses in PU-Learning. Contrastive losses are particularly effective for learning embeddings that are used in retrieval tasks because they directly optimize the relevant ranking metrics. This study proposed a novel debiased CL approach that significantly mitigates the positive bias problem in PU-Learning, enhancing the discriminative power of learned representations.

## III. Proposed Methods

### A. Overview of the Proposed Approach

The general paradigm for semi-supervised semantic segmentation is that the knowledge learned by the teacher network assists in updating the parameters of the student network. The overall framework of our proposed method is shown in Fig. 2, with the overall network divided into two teacher networks and one student network. Labeled images first go through the student network for supervised training, while unlabeled images are augmented separately and passed through the two teacher networks and the student network to generate prediction labels. The prediction labels from the two teacher networks are weighted by confidence, optimizing the training performance from a consistency regularization perspective. This module is named the confidence-weighted dual mean teacher (CDT) module. The predicted labels generated by the teacher and student networks undergo unsupervised training. Unlabeled images and labeled images generate predicted labels through the student network, and queries are selected from the corresponding feature space. Positive keys and hard true negative keys are selected from the feature space corresponding to the predicted features generated by the teacher networks. Finally, biased CL is performed using the queries, positive keys, and hard true negative keys.

The definition of the semi-supervised dataset is as follows:

*Labeled Dataset:* $D_l = \left\{(x_i^l, y_i^l)\right\}_{i=1}^{|D_l|}$

*Unlabeled Dataset:* $D_u = \left\{(x_i^l)\right\}_{i=1}^{|D_u|}$

According to the principle of semi-supervised learning, labeled images represent only a small fraction of all the training images, $|D_l| \ll |D_u|$. The proposed framework consists of a student network and two teacher networks. The student and teacher networks have the same architecture but different parameters, where the parameters $\theta_{t1}$ and $\theta_{t2}$ of the teacher networks are the exponential moving average of the student network parameters $\theta_s$. To differentiate the parameters of teacher networks during initialization, we use unique strategies for each. The first teacher network mirrors the student network in parameter configuration, whereas the second adopts 90% of the first teacher network's parameters. Each network includes an encoder head $h$, a decoder head $f$, and a representation head $r$.

Our method can target the characteristics of remote sensing scenes and perform classification learning at both the image and feature levels, effectively accomplishing semantic segmentation. Combined with modules in Fig. 2, this framework mainly consists of two parts: a consistency regularization module and a loss function based on biased CL. The structure of each part is described as follows.

CDT Module: CDT is a new consistency regularization paradigm for semi-supervised semantic segmentation. Unlike MT, CDT enhances image-level perturbation by combining strong image augmentation and uses two teacher networks for prediction. The predicted labels are weighted by confidence

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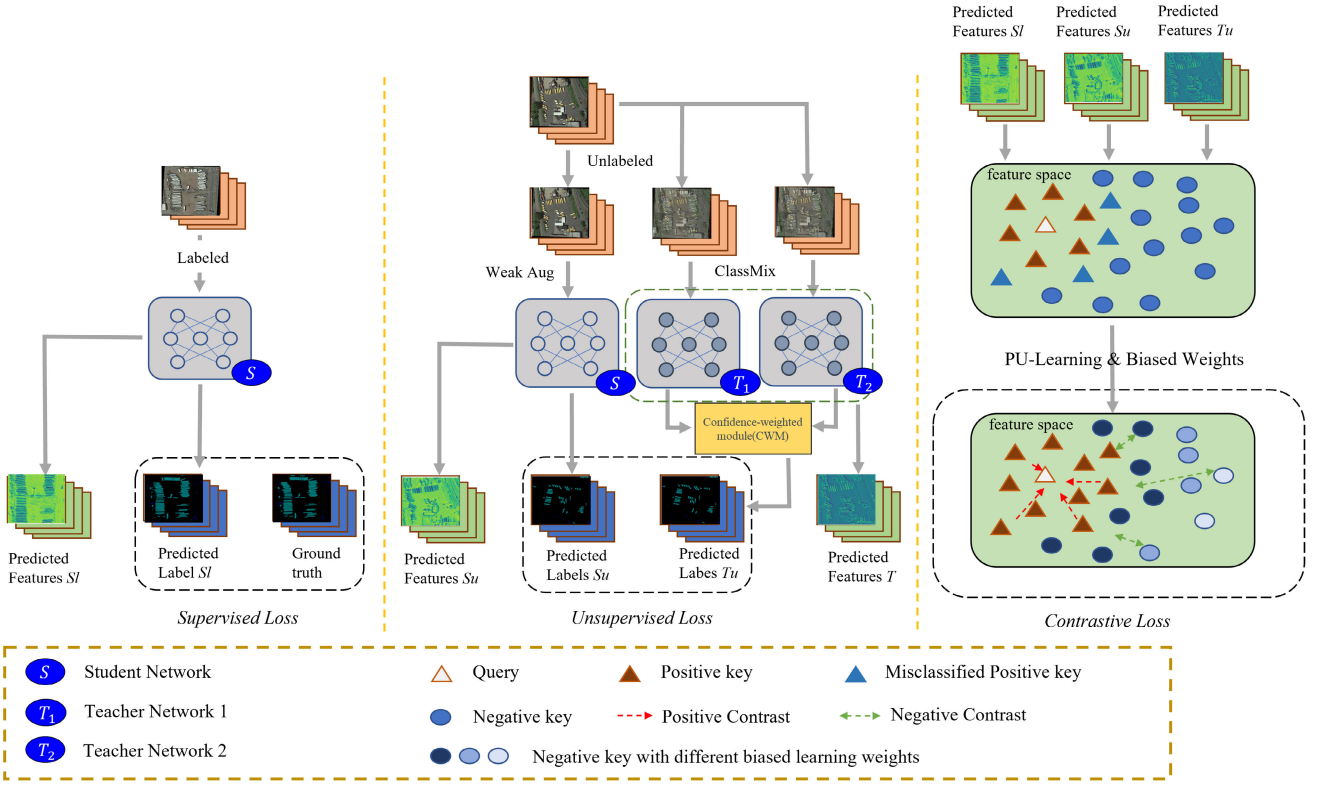



Fig. 2. Overview of the proposed method framework. The framework includes a consistency regularization module, CDT, and a biased CL module. The CDT mainly computes the supervised loss and the unsupervised loss. The supervised loss is computed based on labeled data with ground-truth annotations; the unsupervised loss is computed for unlabeled data with generated pseudolabels from student and teachers. The biased CL module generates the contrastive loss which is computed based on pixel-level dense representation predicted from both labeled and unlabeled images. In feature space, for negative keys, the darker the color, the higher the weight.

to form the final teacher prediction labels, which are used to assist the student network in prediction.

*Biased Contrast Learning (BCL) Module:* BCL acts on the feature level of the image. It first divides the feature space and classifies the predicted pixel-level features into queries, positive samples, and negative samples. The module then uses PU-Learning to filter true negative samples. For negative samples closer to positive samples in the feature space, higher learning weights are assigned, and BCL is performed to better classify features at the feature level.

### B. Confidence-Weighted Dual Mean Teacher Module

In this article, the model consists of two teacher networks and one student network. During each training iteration, the parameters of the student network are refined based on the loss, and the teacher networks' parameters are updated using the exponential moving average of the student network's parameters.

During each training iteration, we carry out a random sampling to obtain an identical number of labeled images $N_l$ and unlabeled images $N_u$, with $|N_l| = |N_u|$. For each labeled image, we input it into the student network to make predictions, and then compare the predicted results with the ground-truth labels to calculate the supervised loss $L_{\text{sup}}$

$$L_{\text{sup}} = \frac{1}{|N_l|} \sum_{(x_i^l, y_i^l) \in N_l} l_{\text{ce}}(y_{si}^l, y_i^l)$$
$$y_{si}^l = \text{onehot}(\text{softmax}(f \circ h(x_i^l; \theta_s))). \quad (1)$$

$l_{\text{ce}}(\bullet)$ represents the cross-entropy loss, and $y_i^l$ denotes the manually annotated labels. $\text{softmax}(f \circ h)$ represents the segmentation probability map generated after the image passes through the encoder $h$ and decoder $f$ successively. $y_{si}^l$ is the final one-hot encoded form. $\theta_s$ denotes the parameters of the student network.

For the unlabeled images, after applying strong image augmentation, the images are fed into the two teacher networks separately to generate different prediction labels. The prediction labels generated by the two teacher networks are combined by confidence-weighted averaging. For the teacher network with a worse performance, a higher weight is given to increase the difference in predicted views to raise the difficulty in maintaining consistency. The weighted averaging of the prediction labels is represented as follows:

$$y_{\text{ti}}^u = \text{onehot}(\text{softmax}(c_1 f \circ h(x_i^u; \theta_{t1}) + c_2 f \circ h(x_i^u; \theta_{t2}))). \quad (2)$$

Here, $c_1$ and $c_2$ are the confidence weights. $\theta_{t1}, \theta_{t2}$ denotes the parameters of the teacher networks. The requirement of consistency regularization is to maintain the consistency of output results while increasing perturbations at various levels. The CDT module mainly studies how to reasonably increase the perturbation at the network level, which needs to keep the network parameters different while the perturbation at the view needs to maintain a reasonable range. Based on this, our model uses $c_1$ and $c_2$ confidence weights, assigning higher learning weights to teacher networks with poorer learning

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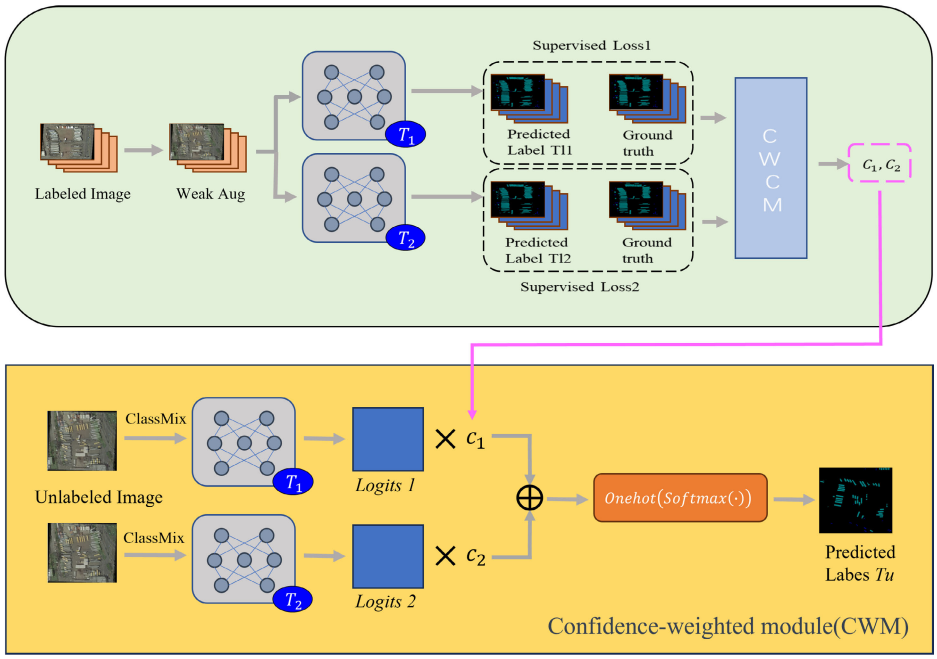



Fig. 3. Computation of network confidence weights and the process of confidence weighting of the prediction labels. First, the computation of confidence weights process integrates the calculation of supervised loss using labeled images following the same procedure. The prediction performances of the two teacher networks are evaluated based on the supervised loss, and the supervised loss generates the confidence weights through the confidence weight calculation module (CWCM). After obtaining $c_1$ and $c_2$, the logits obtained from the teacher network are weighted using the unlabeled image, ultimately yielding the prediction labels.

performance, thereby increasing network-level perturbations. However, fixed confidence weights cannot meet our requirements, as the learning effects of the two teacher networks are random, and we cannot know the specific advantages and disadvantages. Therefore, a dynamic confidence weight needs to be defined for a reasonable evaluation, and this confidence weight must be able to reflect the relative performance of the two teacher networks during each training round. For this reason, we introduce labeled images. The calculation process is shown in Fig. 3. In each training round, while keeping the parameters of the teacher networks unchanged, the labeled images $x^l$ are passed through the teacher networks following the same process for prediction. The generated prediction labels are compared with the ground truth for supervised loss calculation, and $c_1$ and $c_2$ are defined accordingly

$$c_1 = \frac{L^l_{\text{sup1}}}{L^l_{\text{sup1}} + L^l_{\text{sup2}}}$$
$$c_2 = \frac{L^l_{\text{sup2}}}{L^l_{\text{sup1}} + L^l_{\text{sup2}}}. \tag{3}$$

Here, $L^l_{\text{sup1}}$ and $L^l_{\text{sup2}}$ are the supervised losses obtained from labeled images entering the teacher networks for predicting labels, with cross-entropy loss chosen for this purpose. A smaller loss in the teacher network indicates better prediction performance, while a larger loss indicates poorer prediction performance. Therefore, higher confidence weights are given to the corresponding prediction labels. This further strengthens the model's consistency during the learning process.

At the same time, when updating the parameters of the teacher networks, only the parameters of the poorly performing teacher network, i.e., the one with higher confidence weight, are updated.

Unlabeled images $x^u$ also need to enter the student network for prediction, generating a prediction label $y^u_{si}$. This is used for unsupervised learning with the prediction labels $y^u_{\text{ti}}$ generated by the teacher networks, calculating the unsupervised loss, which is defined as follows:

$$L_{\text{unsup}} = \frac{1}{|N_u|} \sum_{(x^u_i) \in N_u} l_{\text{ce}}(y^l_{si}, y^l_{\text{ti}}). \tag{4}$$

### C. Biased CL Module

In Section III-B, we defined supervised and unsupervised losses. Both the losses optimize the prediction performance at the image level. To further optimize learning performance in multiple dimensions, we combined CL based on the image feature space to enhance the feature classification capability.

CL can be understood as a dictionary lookup task [14]. First, define a query keyword and the remaining samples as a dictionary. Samples are divided into positive keys and negative keys, with positive keys generally located at the beginning of the dictionary. The goal of CL is to reduce the difference between the query and the positive key while increasing the difference between the query and the negative key.

Semantic segmentation can be considered as a pixel-level classification task, so we divide the pixels into samples. To achieve this, we added a parallel feature representation decoder head $r$ on top of the additional classification decoder head to provide a higher dimensional feature representation of

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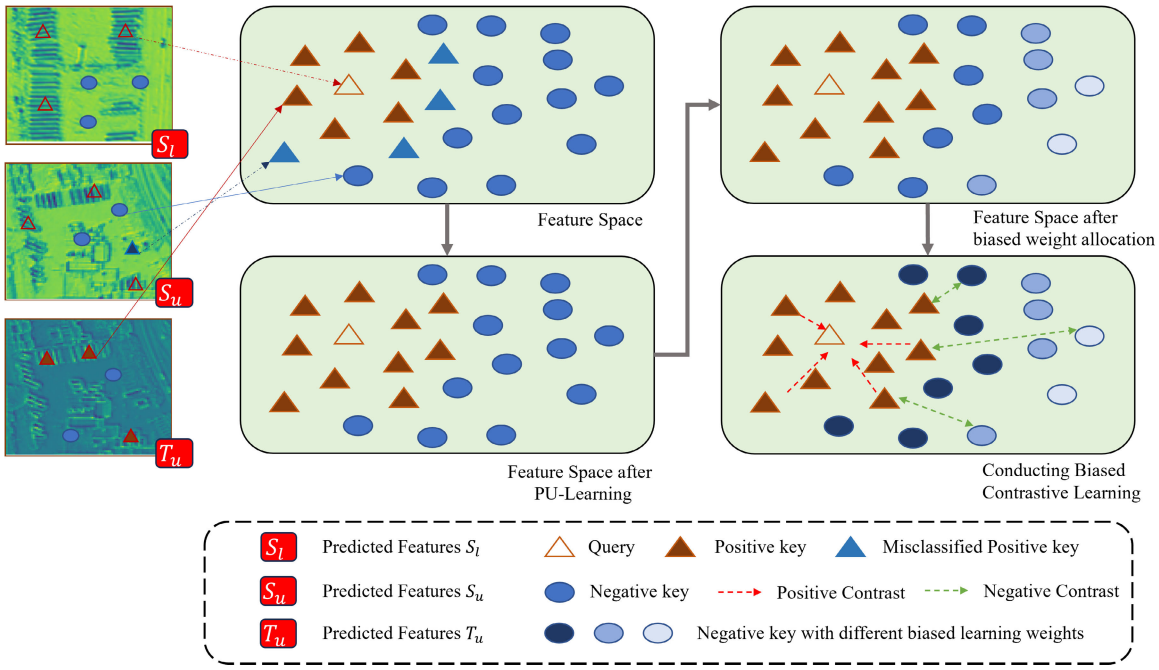



Fig. 4. Flowchart of the biased CL module.

the pixels. All the samples involved in CL are selected from the samples decoded by $r$.

In this article, we use biased CL. The overall process of the biased CL is shown in Fig. 4. In regular CL, the same learning weight is used for different categories of negative learning. However, the reality is that the hard-to-classify negative keys require more learning weights than the easy-to-classify negative keys. Therefore, we use biased CL to assign different bias weights to different categories of negative keys, and define the learning strategy more reasonably. The biased contrastive loss is defined as follows:

$$L_{\text{con}} = -\frac{1}{C \times M}\sum_{c=0}^{C-1}\sum_{i=1}^{M}\log\left[\frac{\lambda_i^p \exp(\text{sim}(z_{ci}, z_{ci}^+)/\tau)}{\lambda_i^p \exp(\text{sim}(z_{ci}, z_{ci}^+)/\tau) + \sum_{j=1}^{J}\lambda_j^n \exp(\text{sim}(z_{ci}, z_{cij}^-)/\tau)}\right]. \tag{5}$$

$C$ represents the aggregate count of segmentation categories, $M$ represents the number of queries and positive keys, and $J$ represents the number of negative keys; sim(•) is a function for calculating similarity; $z = r \circ h(x)$; $\lambda_i^p$ and $\lambda_j^n$ are the biased weights for positive and negative sample pairs, with their specific calculations provided later. In the experiment, we set $M = 50$ and $J = 256$ following [6].

CL is performed on the pixel features of the pseudolabel image, and the overall strategy can be summarized as follows: 1) selecting a query; 2) selecting a positive key; 3) selecting a negative key; and 4) defining asymmetric weights.

*Selecting the Query:* For each class of query, there are two types—queries for labeled images and queries for unlabeled images. The definition of queries for labeled images is as follows:

$$Q_c^l = \mathbb{1}\{y_{ij} = c\}(r \circ h(x_{ij})). \tag{6}$$

For the $i$th labeled image, $x_{ij}$ is the $j$th pixel, and $y_{ij}$ is the annotation of the $j$th pixel. $\mathbb{1}()$ is the indicator function

$$Q_c^u = \mathbb{1}\{\hat{y}_{ij} = c\}(r \circ h(x_{ij})) \tag{7}$$

where $\hat{y}_{ij}$ is the predicted label generated by the student network.

The overall query is

$$Q_c = Q_c^u \cup Q_c^l. \tag{8}$$

Selecting the positive key: Unlabeled images enter the teacher network for prediction, and the positive key is selected from them. For the selection of feature maps from two teacher networks, CL requires rich features. For this reason, we choose the feature maps from the teacher network with better predictive performance for subsequent CL computations. The selection rule is consistent with the query

$$P_c^u = \mathbb{1}\{\tilde{y}_{ij} = c\}(r \circ h(x_{ij})) \tag{9}$$

where $\tilde{y}_{ij}$ is the predicted label generated by the teacher network.

*Selecting the Negative Key:* For unlabeled images, the query and positive key can be easily selected through softmax. As for the negative key, since it is an unlabeled image and has more diverse types compared with the positive key, the common practice is to uniformly sample from the remaining pixel features as the negative key. This is applicable in visual scenes, but for remote sensing images, the problem of intraclass variations and interclass similarities may cause the positive samples of the same class to be not tightly distributed in the feature space, leading to the remaining positive keys being

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sampled as negative samples. This will bring contradictions to the CL training strategy and result in worse training results. To solve this problem, we use PU-Learning to screen the true negative samples. PU-Learning is a learning framework where the training set consists of a small number of positive examples and a large number of unlabeled examples that could be either positive or negative. The goal of PU-Learning is to learn to distinguish between positive and negative examples. According to PU-Learning, the data distribution of positive key and negative key can be divided into

$$\begin{aligned} p(z) &= \alpha^- p(z^-) + \alpha^+ p(z^+) \\ p(z^-) &= (p(z) - \alpha^+ p(z^+))/(1-\alpha^+) \\ \alpha^+ + \alpha^- &= 1. \end{aligned} \tag{10}$$

$p(z)$ represents all the sample features, $p(z^+)$ represents the positive samples of the same class, $p(z^-)$ represents the true negative samples, and $\alpha^+$ and $\alpha^-$ are their respective weights in the feature space. Equation (10) is based on the assumption that the distribution of the entire dataset can be viewed as a mixture of the distributions of positive and negative examples. The weights $\alpha^+$ and $\alpha^-$ in the mixture reflect the proportions of positive and negative examples in the entire dataset. Therefore, by subtracting the distribution of positive examples from the entire data distribution and then dividing by the proportion of negative examples, we obtain the distribution of negative examples.

From this, the true negative samples can be screened. Following [47], the negative sample part of the contrastive loss is updated as

$$\frac{1}{1-\alpha^+}\left(\sum_{j=1}^{J}\exp(\text{sim}(z_{ci}, z_{cij}^-)/\tau) - \frac{J}{M}\alpha^+\sum_{i=1}^{M}\exp(\text{sim}(z_{ci}, z_{ci}^+)/\tau)\right). \tag{11}$$

At the same time, to better distinguish hard-to-classify categories, we adopt biased CL, assigning higher weights to negative samples with higher similarity to positive samples. Since the similarity is used as the evaluation criterion, the biased weight is defined as $\exp(\text{sim}(z_{ci}, z_{cij}^-)/t)$. The final weight update is

$$\begin{aligned} \lambda_i^p &= \frac{M-(M+J)\alpha^+}{M(1-\alpha^+)} \\ \lambda_j^n &= \frac{1}{(1-\alpha^+)}\exp(\text{sim}(z_{ci}, z_{cij}^-)/\tau). \end{aligned} \tag{12}$$

The overall algorithm process of bias CL is shown in Algorithm 1.

**Algorithm 1** Workflow of Biased CL

**Input:** Labeled images $I_l$ and unlabeled images $I_u$
**Output:** Biased Contrastive Loss $L_{con}$

```
Initialize L_c ← 0
for x_i ∈ I_l ∪ I_u do
    Get feature representations: z_i ← r ∘ h(x_i)
    for c ← 0 to C − 1 do
        Get Queries Q_c based on 8;
        Get Positive Keys P_c based on 9;
        Select the true Negative Keys N_c based on 10;
        Update the negative learning part of CL based on 11;
        Calculate the negative learning weights based on 12;
        L ← L + l(Q_c, P_c, N_c) based on 5;
    end
end
Get Biased Contrastive Loss L_con ← L/(M×C)
```

### D. Loss Function

The final loss update of the model in this article is

$$L = L_{\text{sup}} + \lambda_u L_{\text{unsup}} + L_{\text{con}} \tag{13}$$

where $\lambda_u$ is the weight of the unsupervised loss part. As the training progresses, the reliability of the predicted images without labels gradually improves. For the unsupervised loss part, the corresponding learning weight should be gradually reduced. As mentioned above, the misclassified samples when selecting positive and negative samples can also reflect the prediction performance of the network. Therefore, we count the number of all the misclassified positive keys, denoted as $E$

$$\lambda_u = \frac{\sum_{(x_i^u)\in N_u} H_i \times W_i}{\sum_{(x_i^u)\in N_u} H_i \times W_i - E}. \tag{14}$$

$H \times W$ signifies the dimensions of the input unlabeled image, corresponding to the total pixel count.

## IV. Data and Setup

### A. Datasets for Experiment

1) *iSAID*: In this article, we use the iSAID dataset [48] to evaluate the efficacy and performance of our proposed method for semantic segmentation. The iSAID dataset is a large-scale and challenging benchmark in the field of remote sensing, containing a diverse collection of 15 object categories and a total of 2806 high-resolution aerial images. To facilitate the experimental process, the dataset is divided into a training set with 1411 images and a test set comprising 458 images. During the training phase, we adopt a data augmentation strategy that involves randomly cropping the images to a fixed size of $512 \times 512$ pixels.

2) *Vaihingen*: we also use the Vaihingen dataset [49]. The Vaihingen dataset, a benchmark dataset in the realm of remote sensing semantic segmentation, is derived from the Vaihingen region in Germany. It is a collaborative product of the International Society for Photogrammetry and Remote Sensing (ISPRS) and the German Society for Photogrammetry, Remote Sensing, and Geoinformation (DGPF). The dataset comprises high-resolution aerial images with a spatial resolution of 9 cm per pixel, captured using an UltraCamXp large-format digital aerial camera. Each image in the dataset has a dimension of $2494 \times 2064$ pixels, covering an area of approximately $600 \times 600$ m.

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### B. Evaluation Metrics

The assessment measure used here is the mean intersection over union (mIoU). IoU, a commonly adopted metric for tasks of semantic segmentation, gauges the overlap extent between the predicted segments and their respective ground truths. This metric provides a reliable means of assessing the performance of segmentation algorithms, as it takes into account both false positives and false negatives, thereby offering a comprehensive view of the model's accuracy. In the context of semantic segmentation, the IoU is often reported as the mIoU, which is the average IoU across all the classes present in the dataset

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{15}$$

where TP stands for the count of true positives, TN is the number of true negatives, FP symbolizes the quantity of false positives, and FN is the number of false negatives.

### C. Implementation Detail

The network used in the experiment is Deeplab V3+, with ResNet-101 as the backbone. Images are randomly cropped to a size of 512 × 512 before being input into the training network, with a batch size of 1 and training epoch of 200. The learning rate is set to a decayed learning rate $\mathrm{lr}(1 - \mathrm{iter}/\mathrm{iter}_{\max})^{0.9}$. The stochastic gradient descent (SGD) optimizer is used, with an initial learning rate set to 0.01 and a learning rate decay of 0.0005. The dataset is divided into 1/2, 1/4, 1/8, and 1/16 labeled images, with the remaining images being used as unlabeled images for training the model. The strong data augmentation method is classmix with a random pixel ratio of 0.5 for each image. The weight of the auxiliary loss is set to 0.4.

## V. Experimental Results

### A. Experiment on iSAID

*1) Comparison Experiments:* Under the same experimental settings and conditions, we compare the method in this article with the currently widely used and high-performance methods in semi-supervised semantic segmentation. We compare the prediction accuracy for the full dataset, 1/2, 1/4, 1/8, and 1/16 data divisions. The same division ratio is used for all the experiments, with the same data used for training and testing. To reduce the randomness of the results, we conduct each method three times and average the results. The results are shown in Table I.

After analyzing the iSAID dataset, we found that semi-supervised semantic segmentation is highly challenging for certain categories. To demonstrate the segmentation advantages of our method in these categories, we also present the comparison results of the IoU for each category at a 1/4 dataset scale with other methods, as shown in Table II.

From the results, it can be seen that our method has a significant advantage for the extremely difficult-to-segment categories of helicopters and oil tanks. This is because helicopters occupy a very small proportion in all the images, and even if some methods can distinguish helicopters, they tend to misidentify and segment them as airplanes due to their highly similar features. Our method achieves the best performance in most categories.

TABLE I

Comparisons With the SOTA Methods Evaluated on the iSAID Dataset With 1/16, 1/8, 1/4, 1/2, and Full (1) Labeled Data

| Method | 1/16 | 1/8 | 1/4 | 1/2 | full |
|---|---|---|---|---|---|
| MT [5] | 22.13 | 39.76 | 41.91 | 45.33 | 49.97 |
| CutMix [9] | 23.44 | 42.26 | 43.37 | 47.96 | 51.89 |
| RanPaste [50] | 24.71 | 41.11 | 42.38 | 47.06 | 50.29 |
| ICNet [16] | 24.05 | 42.14 | 42.67 | 46.80 | 50.65 |
| GCT [32] | 23.57 | 40.09 | 41.03 | 46.91 | 50.74 |
| PCIS-I [51] | 26.03 | 42.63 | 44.15 | 48.91 | 53.90 |
| **(ours)** | **29.08** | **43.14** | **45.67** | **50.13** | **54.12** |

From Table I, we can see that our method achieved the best results in the cases of 1/16, 1/4, 1/8, 1/2, and full data. Compared with the standard semi-supervised segmentation method, our method achieved a significant improvement in the 1/16 division ratio. The visualization of the segmentation results on the iSAID dataset at a 1/4 dataset scale is shown in Fig. 5

*2) Time Cost of Training and Testing:* To showcase the computational complexity of the framework and the potential scalability issues when applied to large-scale remote sensing datasets, this section will conduct a detailed analysis of the computational efficiency of the framework proposed in this article. Both our method and the comparison methods are carried out in the hardware environment of NVIDIA GeForce RTX 3090, the dataset used is the iSAID dataset with a 1/4 annotation ratio, and the network used is Deeplabv3+. The training and inference times of all the comparison methods are shown in Table III. The inference time of the model on the test set is irrelevant to the framework strategy, and the inference time of all the methods is consistent. For training time, it can be seen that MT has achieved the best time efficiency, and CutMix, Ranpaste, ICNet, and the method of this article have all expanded strategies based on MeanTeacher. In comparison, the performance of the method in this article has achieved the best performance. Under this premise, thanks to the adaptive unsupervised loss weight, the training time of this article is also at a medium level; we believe that it is worthwhile to exchange appropriate inference time for performance improvement.

*3) Ablation Study of Overall Components:* In the proposed semi-supervised semantic segmentation framework, we use unlabeled samples from the perspective of image-level and network-level consistency regularization and confidence. At the same time, the pixel classification at the feature level is optimized. To understand the influence of each component on the method's performance, we progressively ablate each one. The specific ablated components are divided into two categories according to the method.

The consistency regularization part is given below.

1) The CDT network module, which increases network-level disturbances and enhances the model's robustness.

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TABLE II
COMPARISONS WITH THE SOTA METHODS EVALUATED ON THE iSAID DATASET

| Method | SH | RA | BD | TC | BC | GTF | BR | LV | SV | HC | SP | ST | SBF | PL | HA | mIoU(1/4) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MT [5] | 47.53 | **56.29** | 63.27 | 64.79 | 27.84 | 30.32 | 9.03 | 62.49 | 33.68 | 4.43 | 69.40 | **31.17** | 40.57 | 39.84 | 47.99 | 41.91 |
| CutMix [9] | 50.39 | 49.19 | 53.76 | 69.94 | 28.66 | 28.32 | 10.50 | 65.76 | **40.25** | 8.43 | 68.26 | 26.74 | 53.28 | 47.60 | 49.51 | 43.37 |
| RanPaste [50] | 46.63 | 50.58 | 54.98 | 69.35 | 27.99 | 29.39 | 9.36 | **68.12** | 30.21 | 9.14 | 66.44 | 26.68 | 52.15 | 46.20 | 48.44 | 42.38 |
| ICNet [16] | 51.49 | 47.56 | **66.43** | 65.76 | 24.75 | 28.96 | 9.04 | 65.28 | 35.72 | 8.89 | 65.97 | 21.40 | 48.98 | **49.07** | 50.72 | 42.67 |
| GCT [32] | 49.14 | 49.11 | 44.94 | 67.88 | 24.61 | 25.60 | 11.63 | 58.47 | 35.07 | 10.77 | 56.48 | 25.91 | **65.00** | 42.02 | 48.81 | 41.03 |
| PCIS-I [51] | 50.20 | 49.32 | 55.76 | 70.42 | 29.75 | 28.16 | 15.13 | 65.46 | 34.17 | 13.65 | 68.60 | 26.65 | 53.76 | 47.47 | **55.76** | 44.15 |
| **(ours)** | **51.58** | 50.24 | 56.04 | **72.28** | **30.73** | **30.46** | **20.35** | 66.12 | 35.12 | **19.21** | **69.44** | 27.13 | 55.22 | 48.28 | 52.85 | **45.67** |

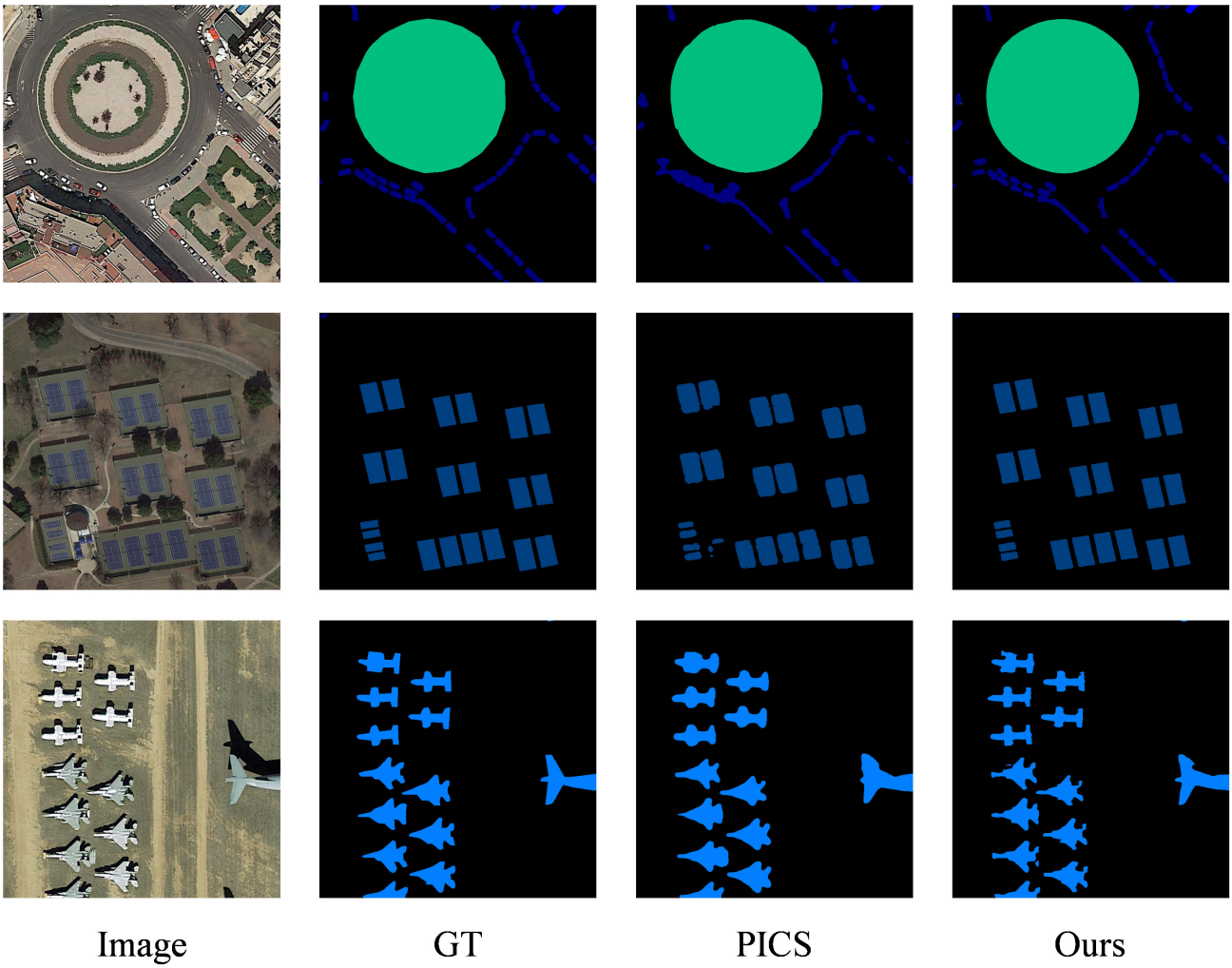



Fig. 5. Visualization results on the iSAID dataset.

2) The image enhancement ClassMix module of the teacher network, which increases image-level disturbances and improves the model's sensitivity to image details.

The CL part is given below.

1) The CL module, which optimizes feature classification and enhances the model's classification performance.
2) The PU-Learning and weighting module, which is used for more accurate feature division before CL and setting the weights for CL.

We conducted the ablation experiments with a dataset split ratio of 1/4 and tested on the iSAID dataset, which is more challenging in the remote sensing field. To ensure the fairness of the experimental results, we performed each experiment three times and took the average.

The experimental results are shown in Table IV. We use the experiment that integrates all the modules as the benchmark. First, all the modules were removed and only the standard consistency regularization model was used for semi-supervised segmentation. The mIoU dropped by 6.57%, indicating the effectiveness of integrating all the modules.

For the ablation experiments of the four modules, the mIoU dropped by 1.98% and 1.84%, respectively, after removing the CDT module and the image enhancement module used

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TABLE III
TIME COST OF TRAINING AND TEST TIMES FOR DIFFERENT METHODS

| Method | Training | Test | mIou(%) |
|---|---|---|---|
| MT [5] | **23.6h** | 2.4min | 41.91 |
| CutMix [9] | 24.3h | 2.4min | 43.37 |
| RanPaste [50] | 28.1h | 2.4min | 42.38 |
| ICNet [16] | 27.4h | 2.4min | 42.67 |
| GCT [32] | 24.5h | 2.4min | 41.03 |
| PCIS-I [51] | 29h | 2.4min | 44.15 |
| **(ours)** | 27h | 2.4min | **45.67** |

TABLE IV
ABLATION STUDY ON THE EFFECTIVENESS OF COMPONENTS IN OUR METHOD

| CL | CDT | Image Aug | PU-learning & Weitht | 1/4 |
|---|---|---|---|---|
| | | | | 39.10 |
| | ✓ | ✓ | | 42.26 |
| ✓ | | ✓ | ✓ | 43.69 |
| ✓ | ✓ | | ✓ | 43.83 |
| ✓ | ✓ | ✓ | | 44.72 |
| ✓ | ✓ | ✓ | ✓ | **45.67** |

TABLE V
ABLATION STUDY ON THE EFFECTIVENESS OF DYNAMIC WEIGHT FOR THE UNSUPERVISED LOSS

| $\lambda_u$ | 0 | 0.75 | 1 | Dynamic | 1.25 | 1.5 |
|---|---|---|---|---|---|---|
| **mIoU(%)** | 43.19 | 44.73 | 45.10 | **45.67** | 45.61 | 45.65 |
| **Train(hours)** | 18 | 24 | 26 | **27** | 27 | 31 |

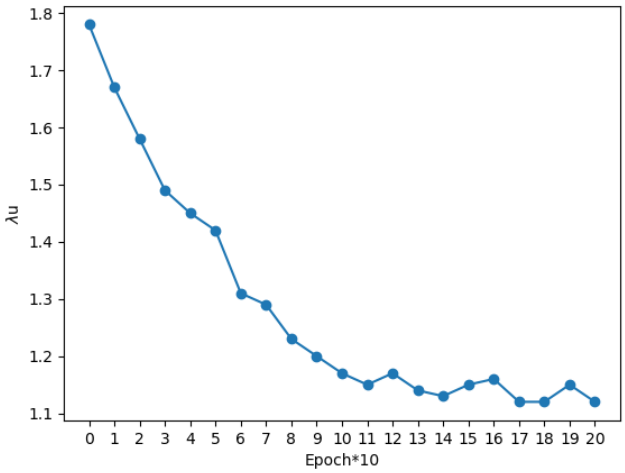



Fig. 6. Variation in dynamic weights $\lambda_u$.

for enhancing consistency disturbances, verifying that adding disturbances at various levels with consistency regularization methods has a notable regularization effect. To validate the effectiveness of the dynamic network confidence weight setting in this article and the rationality of the image enhancement module selection, we conducted more detailed ablation experiments later.

The CL module and the PU-Learning and weight module can effectively enhance the classification ability at the feature level. After removing the CL module, the mIoU dropped by 3.41%, indicating the effectiveness of CL. The PU-Learning and weight module is designed to assist the optimization of feature classification for CL. To explain the role of this part in detail, we also set up specific ablation experiments in the follow-up to visually demonstrate its function.

*4) Ablation Study of Dynamic Weight for the Unsupervised Loss:* This article introduces an innovative dynamic weight approach for the unsupervised loss, aimed at maintaining equilibrium between supervised and unsupervised losses. In the initial phases of model training, a larger unsupervised loss weight amplifies the model's responsiveness to data structure information, facilitating a quicker identification of the appropriate optimization direction. As the training progresses, gradually reducing the unsupervised loss weight allows the model to focus more on supervised data learning, thereby enhancing training stability.

To validate the effectiveness of this dynamic weight, we set various static weights for comparison. The results are presented in Table V.

As shown in Table V, the dynamic weight achieves the best performance in terms of mIoU. Although the performance with an unsupervised loss weight set to 1.5 is close enough to that of the dynamic weight, considering the training time, the dynamic weight is much more efficient than the static weight that $\lambda_u = 1.5$. This is because increasing the weight of the unsupervised loss part increases the learning difficulty and prolongs the training time. The dynamic weight decreases in the later stages of training, reducing the training time. Furthermore, it can be observed that an appropriate reduction in weight in the later stages does not have a negative impact on performance. The variation in dynamic weights is shown in Fig. 6

*5) Ablation Study of CDT:* For the weighted combination of the prediction labels of the two teacher networks, $c_1$ and $c_2$ need to be calculated first, and then the corresponding weights are assigned to each prediction label. To demonstrate the effectiveness and rationality of the confidence weight settings in the model, experiments with different confidence weights are conducted using three settings. These settings are: simple average (Mean), where $c_1$ and $c_2$ are fixed at 0.5; a higher weight is assigned to the teacher network that exhibits better performance (TGH); and a higher weight is assigned to the teacher network that exhibits worse performance (TPH); an extreme case (Extrme), the high weight is set to 0.99, and the low weight is set to 0.01. The experimental results are shown in Table VI. In the prediction results, the teacher network with better performance had the worst effect, with an mIoU 3.11% lower than the original setup. At the same time, the ablation experiment of the comparison module showed that the effect of this weight setting was worse than that of the single teacher network. Giving the prediction label of the better performing teacher network, a higher weight makes the prediction label participating in $L_{\text{unsup}}$ more reliable during each training, and

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TABLE VI
ABLATION STUDY ON THE EFFECTIVENESS OF CDT

| Mean | TGH | Extrme | TPH | 1/4 |
|---|---|---|---|---|
| ✓ | | | | 44.01 |
| | ✓ | | | 42.56 |
| | | ✓ | | 45.06 |
| | | | ✓ | **45.67** |

its consistency with the student network's prediction label is higher. However, this does not conform to the training strategy of consistency regularization and reduces the learning difficulty in the network. Therefore, the consistency model trained in this way cannot well segment the remaining pictures that do not participate in the training.

Compared with the single-teacher network, a simple average of the prediction labels improves the mIoU by 0.32%. Similar to the single-teacher network, the simple average also brings adverse disturbances to the prediction label, which is beneficial for consistency regularization. But compared with TPH, the simple average has an mIoU 1.66% lower. This indicates that increasing the unreliability of the prediction label and enhancing the difference in views effectively improves the training performance of the network. Of course, the difference in views must be reasonable and cannot be blindly increased. In setting Extrme, we increased the unreliability of the prediction label, but the mIoU was not as good as our dynamic weight setting.

*6) Ablation Study of Image Augmentation:* To gauge the quantitative influence of image augmentation on our approach, we conducted experiments with strong augmentation, weak augmentation, and no augmentation for the two teacher networks, all while keeping all other module settings constant. The strong augmentations chosen were ClassMix and CutMix, while the weak augmentations were flip and scale. The results are shown in Table VII.

Using the no-augmentation setting as a baseline, the mIoU increased by 0.12% with weak augmentation. This minor enhancement already demonstrates the effectiveness of augmentation. However, the effects of strong augmentation were more significant. ClassMix and CutMix increased the mIoU by 1.84% and 1.6%, respectively, indicating that increasing the viewpoint-level variations while maintaining model consistency can significantly enhance model performance.

Comparing the two strong augmentation strategies, ClassMix and CutMix, we finally chose ClassMix as our augmentation strategy for the following reasons.

1) ClassMix places more emphasis on class balance compared with CutMix. CutMix mainly augments data by randomly mixing images, which can introduce class-imbalance issues. ClassMix, however, takes into account the balance of different classes during image mixing. This prevents certain classes from becoming dominant, thereby making the model's recognition abilities more balanced across all the classes. Moreover, in remote sensing images, images of the same class may have a great deal of internal variation (intraclass variations), while images of different classes may have similar features (interclass similarities). ClassMix, compared with CutMix, is better equipped to handle this situation. By mixing images with ClassMix, more intraclass variations and interclass similarities are introduced while preserving the original class information, thus enhancing the model's generalization ability. In terms of experimental results, the average mIoU of three experiments using ClassMix was 0.24% higher than CutMix, verifying the superiority of the theory.
2) ClassMix, through its innovative mixing strategy, does more than simply mix images. It mixes features from different classes. This mixing strategy allows the model to encounter a broader range of feature combinations during the learning process, leading to more rich feature learning. This is particularly beneficial to the subsequent biased CL module in our method.

TABLE VII
ABLATION STUDY ON THE EFFECTIVENESS OF AUGMENTATION

| No Aug | Weak Aug | Strong Aug | 1/4 |
|---|---|---|---|
| ✓ | | | 43.83 |
| | ✓ | | 43.95 |
| | | ✓ (CutMix) | 45.43 |
| | | ✓ (ClassMix) | 45.67 |

TABLE VIII
QUANTITATIVE ANALYSIS OF PU-LEARNING'S EFFECTIVENESS

| Category | SP | PL | SV | RA | LV | HL |
|---|---|---|---|---|---|---|
| Original | 61 | 55.7 | 69 | 49.7 | 71.1 | 9.7 |
| PU-Leaning | 82.6 | 64.8 | 83.2 | 62.3 | 79.5 | 36.5 |

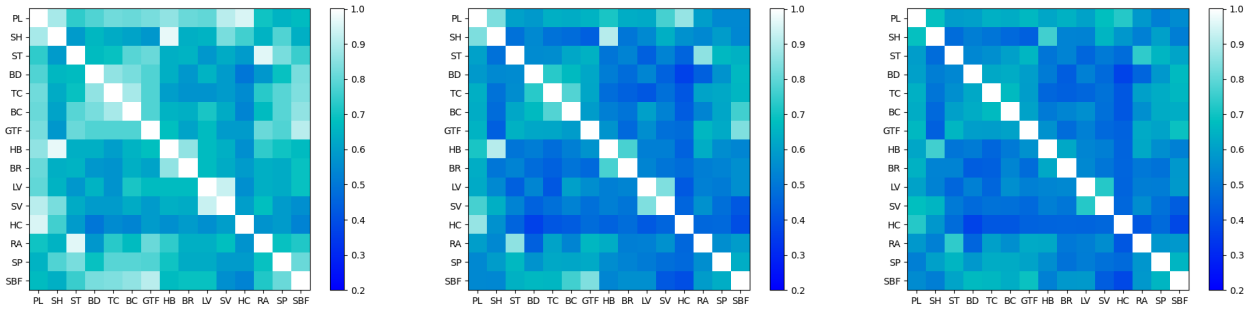
TABLE IX
COMPARISONS WITH THE SOTA METHODS EVALUATED ON THE VAIHINGEN DATASET WITH 1/8, 1/4, AND 1/2 LABELED DATA

| Method | 1/8 | 1/4 | 1/2 |
|---|---|---|---|
| MT [5] | 71.94 | 76.54 | 78.07 |
| CutMix [9] | 73.05 | 77.27 | 79.29 |
| RanPaste [50] | 72.11 | 77.69 | 79.51 |
| ICNet [16] | 73.25 | 77.17 | 78.93 |
| GCT [32] | 71.34 | 77.73 | 79.11 |
| PCIS-I [51] | 74.75 | 78.58 | 79.59 |
| **(ours)** | **75.93** | **79.01** | **80.09** |

*7) Ablation Study of Biased CL:* CL optimizes label prediction at the image feature level. To better demonstrate the effectiveness of CL and the PU-Learning-based weight module, we compare the final results through visualization and feature-level category similarity.

We use the similarity of semantic categories in feature space as an evaluation metric, with color intensity representing the visualization of high similarity. This can better help us display the changes in each category with other categories in the feature space. The experimental settings are: 1) not using CL; 2) using standard CL; and 3) using biased CL.

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(a) (b) (c)

Fig. 7. Visualization of semantic class relation graph. (a) Similarity heatmap for each semantic category pair without using CL. (b) Similarity heatmap following the application of CL. (c) Similarity heatmap after the utilization of biased CL.

TABLE X
COMPARISONS WITH THE SOTA METHODS EVALUATED ON THE VAIHINGEN DATASET

| Method | Imp.surf. | Buildings | Low veg. | Tree | Car | mIoU(1/4) |
|---|---|---|---|---|---|---|
| MT [5] | 86.21 | 91.30 | 70.48 | 64.89 | 69.82 | 76.54 |
| CutMix [9] | 87.68 | 90.24 | 71.74 | 65.41 | 71.28 | 77.27 |
| RanPaste [50] | 88.57 | 90.69 | 71.44 | 66.75 | 71.00 | 77.69 |
| ICNet [16] | 86.20 | 91.43 | 70.04 | 66.01 | 71.17 | 77.17 |
| GCT [32] | 86.04 | **91.85** | 70.68 | 67.47 | 72.61 | 77.73 |
| PCIS-I [51] | 88.80 | 90.37 | 73.66 | 66.88 | 73.19 | 78.58 |
| **(ours)** | **89.81** | 91.06 | **72.58** | **68.18** | **73.42** | **79.01** |

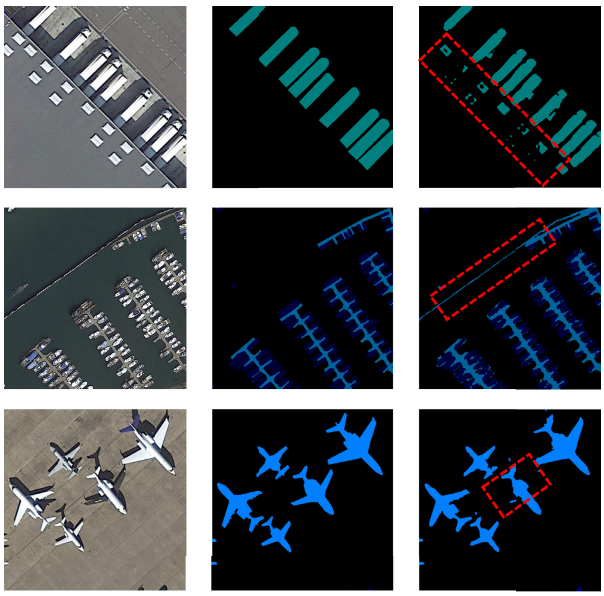


Image GT Ours

Fig. 8. Visualization results of failure cases.

As shown in Fig. 7, without using contrastive loss in the experiment, the final similarity of each feature is relatively high. After using standard CL, the similarity of some easily distinguishable category pairs, such as plane and tennis court, has significantly decreased. These category pairs are already distant in feature space, so traditional CL can optimize the classification of these classes from the feature level. However, the similarity of some difficult-to-distinguish category pairs, such as harbor and ship, has not decreased significantly. The similarity graph of category pairs after using biased CL is shown in Fig. 7(c). The learning effect of easy to distinguish category pairs is basically the same as that of the traditional CL, but the learning effect of difficult to distinguish, that is, category pairs with close distribution in the feature space, is more significant.

*8) Quantitative Analysis of PU-Learning's Effectiveness:* In this section, we evaluate and demonstrate the effect of PU-Learning. To more intuitively and accurately analyze the role of PU-Learning, we select ten annotated images, input them into the teacher network to obtain predicted feature maps, and allocate positive keys and negative keys. We then perform PU-Learning, reassign the positive keys and negative keys, and compare them with the ground truth, using accuracy (%) as the evaluation metric. The results are shown in Table VIII.

*9) Analysis of Failure Cases:* Although our method has demonstrated superior performance across the overall dataset, there are still areas where it falls short, as specifically illustrated in Fig 8. Due to our approach of assigning higher weights to more challenging samples, simpler ones, such as LV and HA, do not receive as much emphasis during classification. This can lead to their misclassification or confusion with certain backgrounds. In addition, our method faces challenges in segmenting some samples due to issues with color and shadows. This issue may stem from a lack of shadow-class sample data or limitations in the training network. Consequently, this calls for appropriate data augmentation or modifications to the network module to address these deficiencies.

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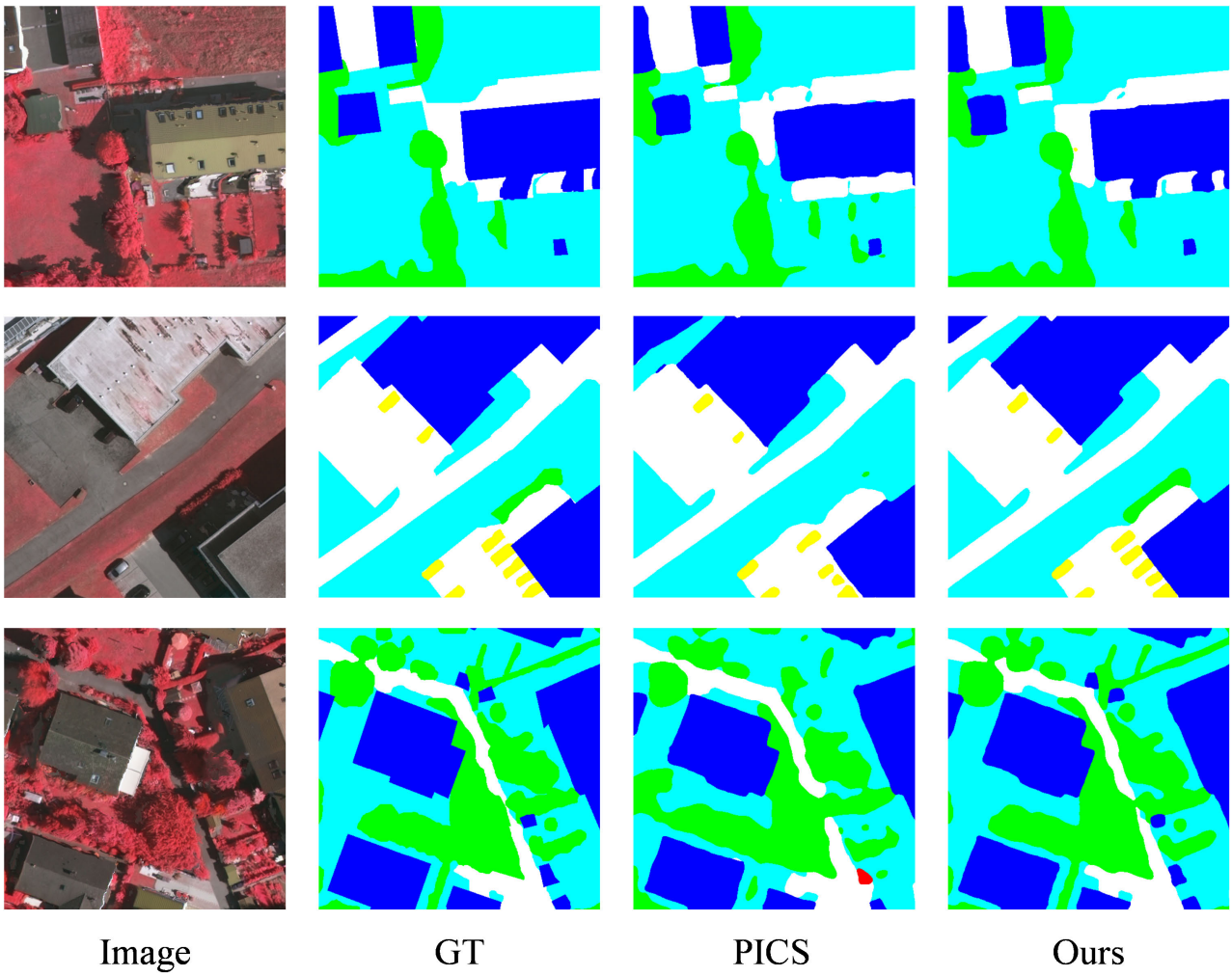



Fig. 9. Visualization results on the Vaihingen dataset.

### B. Experiment on Vaihingen

To validate the applicability of our method in different remote sensing scenarios, we also conduct semi-supervised semantic segmentation experiments on the Vaihingen dataset. The dataset is divided into proportions of 1/8, 1/4, and 1/2. The comparison with other methods is shown in Table IX, where it can be seen that our method achieves the best results in terms of the mIoU evaluation metric. Similarly, to analyze the segmentation performance for specific categories, we list the IoU values for individual categories at the 1/4 dataset proportion and compare them with other methods. As shown in Table X, for five specific categories, our method has achieved an mIoU score of 89.81%, 68.18%, 73.42%, and 72.58% for the four foreground categories: Impervious Surfaces, Trees, Cars, and Low Vegetation, respectively. These results demonstrate a significant improvement over the comparison methods for these four categories. The experiment conducted on the Vaihingen dataset verifies that our approach is applicable to various types of remote sensing targets, exhibiting stable and efficient performance. The visualization of the segmentation results on the Vaihingen dataset is shown in Fig. 9.

## VI. Conclusion

In this article, we focus on the intraclass variations and interclass similarities inherent in remote sensing images and propose a new semi-supervised semantic segmentation framework. This framework combines the methods of consistency regularization and CL techniques to boost the performance of semi-supervised semantic segmentation. Initially, we establish a unique learning paradigm for consistency regularization, integrating the predictive results of two teacher networks with confidence weights and asynchronously updating the teacher network parameters, thereby increasing perturbations at the network level. Second, regarding the feature space of the images, we introduce biased CL to achieve a better partition and classification. By leveraging the biased weights from PU-Learning and positive–negative learning components, we improve the classification ability of the feature space and reduce the classification errors caused by feature sampling errors and hard to classify samples. Our method outperforms SOTA on the Vaihingen and iSAID datasets. In forthcoming studies, we aim to transcend the boundaries of conventional semi-supervised learning. Our aspiration is to adapt the open-world semi-supervised approach for application in the field of

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remote sensing, thereby achieving generalization in segmenting unknown classes. At the same time, we are conducting further research into issues such as shadows affecting our current targets, which lead to segmentation failures, to achieve more superior segmentation results.

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